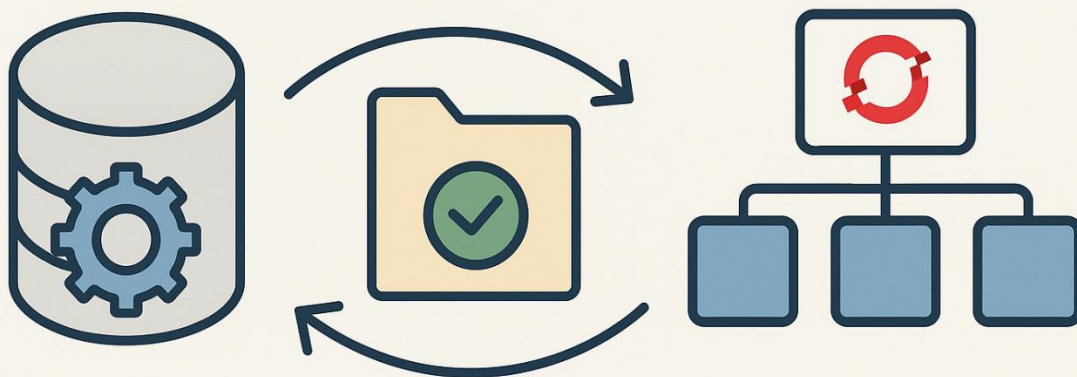


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ETCD Backup and Restore in OpenShift

ETCD BACKUP AND RESTORE IN OPENSIFT



Introduction to ETCD in OpenShift

What is ETCD?

ETCD is a distributed key-value store used by OpenShift (and Kubernetes) to **store the complete state of the cluster**. Every configuration, status, and metadata of the OpenShift cluster — such as pods, nodes, services, secrets, roles, and policies — is stored inside etcd.

Why is ETCD So Important?

In OpenShift, the control plane components interact with etcd to make cluster-wide decisions. If etcd is corrupted or lost, your OpenShift cluster **cannot function properly**, and most critical services will **fail or disappear**.

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ETCD is responsible for:

- Maintaining cluster state and configuration.
- Acting as a single source of truth for the control plane.
- Supporting fault tolerance and high availability through etcd clustering.

Real-world Example:

If a developer creates a pod or deployment, that information is written to etcd. If the system crashes and etcd is not backed up, **that pod's configuration is lost** — even if the underlying compute nodes are still running.

That's why **regular backup of etcd** is one of the most important tasks for any OpenShift administrator.

Why ETCD Backup is Critical

What Happens Without Backup?

- You lose **all custom resources** and configurations.
- Troubleshooting or rebuilding the cluster becomes almost impossible.
- Application downtime and data inconsistency become unavoidable.

Backup Use Cases:

1. **Disaster Recovery:** Helps restore your cluster in case of node or etcd failure.
2. **Upgrade Rollback:** Useful when major OpenShift upgrades go wrong.
3. **Cluster Migration:** When moving from one data center/cloud to another.
4. **Testing:** Use backups to spin up staging/test environments identical to production.

What Does a Backup Include?

When you perform a proper etcd backup in OpenShift, it includes:

- Cluster state (Pods, Deployments, Services)
- Cluster-scoped resources (CRDs, Roles, Namespaces)
- Static pod manifests
- All Kubernetes control plane configurations

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However, etcd backups **do not include**:

- Actual container image files (stored separately)
- Logs
- Data stored in persistent volumes (should be backed up through CSI snapshots or volume plugins)

Key Notes:

- etcd is version-sensitive. Always restore into the same OpenShift version or take care of version compatibility.
 - Backup files must be safely stored and verified regularly.
-

How to Perform ETCD Backup in OpenShift

Pre-Requisites:

- Admin access to an OpenShift control plane node.
- Sufficient disk space in the backup directory.
- Ensure cluster is in a healthy state (oc get co shows all green).

Backup Steps (OpenShift 4.x):

Step 1: Login to Control Plane Node

```
ssh core@<master-node-ip>
```

Step 2: Run Built-in Backup Script

OpenShift provides a ready-to-use backup script:

```
sudo -E /usr/local/bin/cluster-backup.sh /home/core/backup
```

What this command does:

- Takes a snapshot of etcd (snapshot.db)
- Collects static pod definitions and kube manifests
- Saves everything in the given backup directory

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Sample Backup Output Structure:

```
/home/core/backup
├── etcd
│   └── snapshot.db
├── static_kubernetes
│   ├── kube-apiserver-pod.yaml
│   ├── kube-controller-manager-pod.yaml
│   └── etcd-pod.yaml
└── version
```

Step 3: Copy to a Secure Location

Transfer the backup to a storage server or cloud bucket:

```
scp -r /home/core/backup user@backup-server:/data/backups/
```

Tips:

- Use cron jobs to automate this script.
- Always verify backup size and content.
- Maintain retention for 7 to 30 days depending on policy.

How to Restore ETCD in OpenShift

When Do You Restore?

- etcd cluster is unhealthy or unreachable
- Control plane failure
- Cluster-wide accidental misconfiguration
- Migration or disaster simulation

Types of Restore:

1. **Disaster Recovery Restore:** Entire etcd cluster needs rebuilding.
2. **Single-node Restore (Non-HA):** Useful in lab environments.

Steps for Disaster Recovery Restore:

Step 1: Prepare the Node

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- Select a master/control-plane node.
- Ensure no etcd is running (ps aux | grep etcd)
- Move to the backup location:

```
cd /home/core/backup
```

Step 2: Run Restore Script

```
sudo -E /usr/local/bin/cluster-restore.sh /home/core/backup
```

This script:

- Wipes old etcd data
- Rebuilds etcd directory with snapshot
- Replaces static pod definitions

Step 3: Restart Node

After successful restore, reboot the node:

```
sudo reboot
```

Step 4: Validate Cluster Health

```
oc get nodes
oc get pods -A
oc get co
```

Check that all pods and controllers are running correctly.

Additional Notes:

- For HA setups, restore on one node, then reconfigure and rejoin other control-plane nodes.
- The restore must be done **carefully** with correct backup path and version matching.

Best Practices & Final Thoughts

Best Practices:

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1. Automate Backups

- Schedule backups using systemd timers or cron.
- Always test the automation and validate output.

2. Secure Backup Files

- Use encrypted file systems or tools like gpg.
- Store backups off-cluster — cloud, S3, external drives.

3. Test Restores

- Regularly test restore on a dev or staging cluster.
- Document and simulate DR scenarios quarterly.

4. Monitor ETCD Health

- Regularly check etcd member health:

```
oc exec -n openshift-etcd etcd-<name> -- etcdctl endpoint health
```

- Watch for disk latency and space issues.

5. Track OpenShift Versions

- Do not restore an etcd snapshot taken from a different OpenShift version.
- Take a backup before every OpenShift upgrade.

Summary:

- ETCD is the **single source of truth** for your OpenShift cluster.
- **Backing it up** ensures you don't lose critical data.
- **Restoring it** correctly brings your cluster back to life after a failure.
- Learn the commands, test frequently, and always document your recovery steps.

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